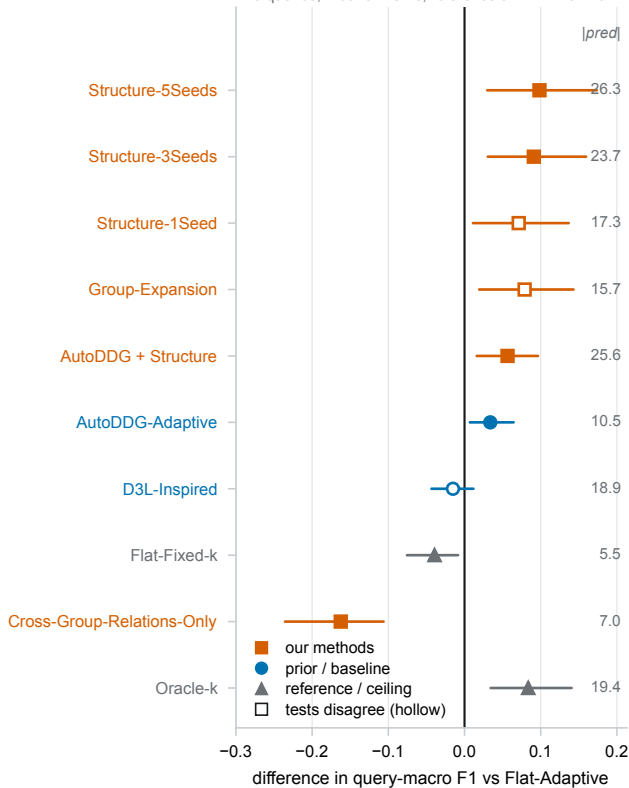


A Comprehensive retrieval comparison

245 queries, 2 benchmarks, reference arm F1 = 0.270



B Component ablation and controls

each contrast is paired and uses its own stated control

Group-Expansion vs flat @ same size
 Structure-1Seed vs flat @ same size
 Structure-3Seeds vs flat @ same size
 drop Group-Expansion
 add Join Candidates
 add Cross-Group Hop
 seeds: 3 vs 1
 seeds: 5 vs 3
 question gate: off vs on
 budget: 20 vs 40

difference in query-macro F1 vs the stated control

C Query-level effect heterogeneity

Structure-3Seeds vs Flat-Adaptive

